Threat Learning, Fear Conditioning, and Traumatic Memory

Chapter 10

In this chapter

Learning objectives 2
Introduction 2
10.1 Pavlovian conditioning 2
10.2 Unconditioned and conditioned responses 3
10.3 Acquisition of threat learning 3
10.4 Context conditioning 4
10.5 Fear generalisation 4
10.6 Avoidance learning 4
10.7 Extinction and renewal 5
10.8 Prediction error 5
10.9 Intrusive memory 6
10.10 Clinical translation of learning science 6
10.11 Safety learning and inhibition 6
10.12 Beyond fear: disgust, shame and moral injury 7
Synthesis 8
Composite case study 8
Review and discussion questions 9
Research assignments 9
Chapter summary 9
References 9

Central question. How does a neutral event become a danger signal, and why does the learning sometimes fail to extinguish?

Learning objectives

- explain the basic elements of Pavlovian conditioning;
- describe acquisition, generalisation and contextual control of fear;
- distinguish extinction, habituation and habituation-like phenomena;
- explain prediction error and its role in updating threat beliefs;
- evaluate the translation of learning science into clinical practice.

Introduction

Fear conditioning is the best-understood model of how trauma-related learning occurs. A neutral stimulus is paired with an aversive event; the neutral stimulus acquires the capacity to elicit defensive responses; and the learning can persist long after the aversive event has ended. The model is powerful because it is experimentally tractable, conserved across species and grounded in identified circuits.

It is also incomplete. Human trauma rarely consists of a single, well-timed pairing in a laboratory. It involves complex events, uncertain threat, social meaning and repeated exposure. The translation from conditioning experiments to clinical trauma is therefore indirect, and the chapter is careful to mark where the model illuminates and where it misleads.

> **A note on fear versus anxiety.** *Fear is typically described as a response to imminent, identifiable threat; anxiety as a response to uncertain, anticipated threat. The distinction is useful clinically but not absolute, and the underlying circuits overlap. Much of what trauma survivors describe is closer to sustained anxiety than to phasic fear.*

10.1 Pavlovian conditioning

Pavlovian conditioning pairs a conditioned stimulus (CS), initially neutral, with an unconditioned stimulus (US) that elicits an unconditioned response (UR). After repeated pairings, the CS elicits a conditioned response (CR). In fear conditioning, the US is typically an aversive shock or loud noise, and the CR includes freezing, autonomic arousal and potentiated startle.

The conditioning preparation allows precise control of stimulus timing, intensity and contingency. This control is why it has been so productive: researchers can manipulate variables and measure outcomes with rigour. It is also why findings do not map neatly onto real-world trauma, where contingencies are ambiguous and exposures are uncontrolled.

10.2 Unconditioned and conditioned responses

The unconditioned response to threat is organised by brainstem and hypothalamic circuits. It includes cardiovascular changes, respiration, analgesia, freezing and endocrine activation. These responses are adaptive in the presence of danger and can be elicited without cortical processing, which is why they can occur before a person has consciously identified the threat.

The conditioned response is acquired through learning and is more flexible. It can be modulated by context, expectation and prior experience. In humans, conditioning can occur with subliminal stimuli, which shows that awareness is not always necessary—though the interpretation of such findings is debated.

Conditioned responses are not a single thing. Different measures—skin conductance, startle, freezing, heart rate—can dissociate. A person may show physiological conditioning without reporting fear, or report fear without physiological change. This matters for assessment.

10.3 Acquisition of threat learning

Acquisition depends on contingency, contiguity and salience. The CS must predict the US; the closer in time, the stronger the learning, though there are exceptions. The intensity of the US matters: more intense aversive events produce stronger conditioning.

At the circuit level, sensory information about the CS reaches the lateral amygdala from the thalamus and cortex. The US information also reaches the amygdala. Convergence of the two inputs produces synaptic plasticity, so that the CS alone can later activate the central nucleus and produce defensive responses (LeDoux, 2000; Maren, 2001).

Acquisition is modulated by stress hormones. Noradrenaline and glucocorticoids enhance consolidation of fear learning, which is adaptive in danger but can produce overly strong memories when stress is extreme.

10.4 Context conditioning

Contexts—the places, times and internal states in which an event occurs—become associated with threat through a process that depends on the hippocampus and amygdala. Contextual fear is often more diffuse than cue-specific fear: a person may feel unsafe without knowing why.

Context processing is relevant to PTSD, where symptoms often fluctuate with setting. A survivor may feel relatively calm at home and overwhelmed in a crowd, not because the crowd is inherently dangerous but because it resembles the context of the event. Hippocampal dysfunction may impair the ability to distinguish safe from dangerous contexts.

Context also matters for extinction. Extinguished fear can return when the context changes—renewal—which is why exposure therapy should be conducted across multiple contexts rather than in a single clinic room.

10.5 Fear generalisation

Generalisation is the tendency to respond to stimuli that resemble the CS. It is adaptive: a person who has been attacked by one dog may reasonably be wary of others. It becomes maladaptive when generalisation is excessive, so that many neutral situations elicit fear.

Generalisation gradients depend on perceptual similarity, prior experience and anxiety. People with PTSD and anxiety disorders often show broader generalisation than controls, and this has been studied with conditioning paradigms in which participants are trained to discriminate between a threat cue and a safety cue.

The clinical implication is that exposure should include discrimination training: helping the person learn not only that the feared situation is safe but that similar situations can also be safe. This is more demanding than simple habituation.

10.6 Avoidance learning

Avoidance is a learned response that prevents the US from occurring. It is negatively reinforced because it reduces anxiety, which makes it resistant to extinction. Avoidance can be passive (freezing, not going out) or active (leaving, checking, reassurance-seeking).

At the circuit level, active avoidance depends on amygdala–striatal pathways, while passive freezing depends on amygdala–brainstem

pathways (Fenster et al., 2018). The distinction matters because the two patterns may respond to different interventions. A patient who freezes may need graded exposure; a patient who flees may need to learn that staying is tolerable.

Avoidance is central to the maintenance of PTSD. Each avoidance prevents the disconfirmation of the threat prediction, so the belief remains unchanged. Treatment therefore requires reducing avoidance in a planned, tolerable way.

10.7 Extinction and renewal

Extinction is the reduction of the conditioned response when the CS is repeatedly presented without the US. It is new learning, not erasure, and it depends on the ventromedial prefrontal cortex and the amygdala (Milad & Quirk, 2012).

Extinction is context-dependent. Renewal occurs when the CS is tested in a context different from the extinction context. Reinstatement occurs after an unsignalled aversive event. Spontaneous recovery occurs with the passage of time. These phenomena explain relapse after apparently successful treatment.

Clinical translation requires attention to context. Exposure should be conducted in multiple settings, at different times, and with varied stimuli. Relapse should be anticipated and normalised rather than treated as failure.

10.8 Prediction error

Prediction error is the difference between what is expected and what occurs. Learning is driven by prediction error: if the US is fully predicted, little new learning occurs; if it is surprising, learning is stronger.

This principle has practical implications. Exposure that is conducted in a way that confirms the person's expectation ("I knew this would be awful") produces little learning. Exposure that violates the expectation ("I expected to panic and I did not") produces more. Contemporary exposure therapy therefore emphasises expectancy violation rather than mere repetition (Craske et al., 2014).

Prediction error also explains why safety behaviours can undermine treatment: if a person only manages because they carry a water bottle,

they may attribute safety to the bottle rather than learning that the situation is tolerable.

10.9 Intrusive memory

Intrusive memories are involuntary recollections that intrude into awareness. They can be sensory (images, sounds, smells), emotional or somatic. They differ from ordinary memories in their vividness, their sense of presentness and their resistance to voluntary control.

Several mechanisms have been proposed: strong perceptual encoding, impaired contextual binding, and failure of prefrontal inhibition. The evidence supports a role for each, but no single account explains all intrusions. Intrusions can occur in people without PTSD, and not everyone with PTSD has them.

Treatment approaches include trauma-focused cognitive therapy, imagery rescripting and eye movement desensitisation and reprocessing. The evidence supports trauma-focused therapies generally; debates about specific mechanisms are less settled.

Cognitive models emphasise the role of negative appraisals and of poor contextual integration in the persistence of intrusions (Ehlers & Clark, 2000).

10.10 Clinical translation of learning science

Learning science has influenced clinical practice in several ways: exposure therapy, extinction-based models, prediction-error accounts and reconsolidation-based interventions. It has also produced cautionary findings: extinction is not erasure, relapse is normal, and context matters.

The translation is not one-to-one. Clinical exposure is more complex than laboratory extinction: it involves meaning, relationship, safety and the person's goals. A good clinician uses learning principles without reducing the person to a conditioning preparation.

10.11 Safety learning and inhibition

Fear learning has a counterpart: safety learning. A cue that predicts the absence of danger acquires the capacity to reduce fear, and discrimination between threat and safety cues is fundamental to adaptive functioning.

Safety signals differ from extinction. Extinction involves the omission of an expected aversive outcome for a cue that previously predicted it; safety learning involves a distinct cue that predicts safety. In experimental paradigms, safety cues reduce fear and cortisol responses, and they engage ventromedial prefrontal and hippocampal circuits.

In PTSD, discrimination between threat and safety cues is often impaired. A person may respond as if everything is dangerous, which is exhausting and self-reinforcing: overgeneralisation leads to avoidance, avoidance narrows experience, and narrower experience provides fewer opportunities to learn safety.

Treatment implications are practical. Exposure should include explicit discrimination—this situation is safe, this one is uncertain, this one is dangerous—rather than global reassurance. Safety behaviours, however, are not the same as safety learning: carrying a water bottle may reduce anxiety without teaching anything, whereas systematic experience of safety with the behaviour reduced provides new information.

The distinction also matters for services. A clinic that feels unsafe, a waiting room that is chaotic, or a clinician who is unpredictable undermines safety learning regardless of the therapy protocol.

10.12 Beyond fear: disgust, shame and moral injury

Fear is not the only emotion organised by learning, and it is not always the central one in trauma.

Disgust is acquired through associative learning and is more resistant to extinction than fear. It features in sexual trauma, in experiences of contamination and in some forms of self-directed revulsion. Exposure that works for fear may need adaptation for disgust, because the response is not easily reduced by repeated contact alone.

Shame is a self-conscious emotion involving global negative evaluation of the self. It is common after interpersonal trauma, particularly childhood abuse, and it predicts depression, self-harm and social withdrawal. Shame is maintained by concealment and by imagined or actual rejection; interventions that address it directly—compassion-focused approaches, trauma-focused cognitive therapy—may be more useful than exposure alone.

Moral injury describes enduring distress following actions or experiences that violate deeply held moral expectations: perpetrating

harm, witnessing injustice, failing to prevent harm, or being betrayed by trusted authorities (Litz et al., 2009). The concept originated in military contexts and has been extended to other settings. It overlaps with PTSD and with shame, guilt and anger, and its distinctiveness is debated.

These phenomena should not be bolted on to fear models and treated as variants of conditioning. They require formulations that include meaning, identity and value. A veteran who feels contaminated by what he did is not simply exhibiting a conditioned response, and treating him as though he were would miss the point.

Synthesis

Table 10.1 summarises conditioning phenomena and clinical analogues.

Table 10.1. Learning phenomena and clinical parallels

Phenomenon	Laboratory description	Clinical parallel	Implication
Acquisition	CS paired with US	Event followed by danger	Threat learning is rapid
Generalisation	Response to similar cues	Fear of many situations	Discrimination training
Context conditioning	Hippocampus-dependent	Setting-dependent symptoms	Vary exposure contexts
Extinction	CS without US	Exposure therapy	New learning, not erasure
Renewal	Context change	Relapse after treatment	Multi-context practice
Reinstatement	Unsigned aversive event	Relapse after stress	Plan for recurrence
Avoidance	Prevented US	Safety behaviours	Reduce gradually

Composite case study

A patient avoids buses after an assault. Treatment begins with imaginal exposure to the memory, then graded use of buses, then travel at different times and routes. Safety behaviours are reduced gradually. The patient is told that anxiety may return after a stressful event and that this is a normal feature of extinction, not a sign of failure.

Review and discussion questions

Why is extinction not the same as forgetting?

What are the clinical implications of renewal and reinstatement?

How does prediction error change the way exposure should be conducted?

Why can avoidance persist even when the person knows it is unhelpful?

What are the limits of the conditioning model for understanding human trauma?

How should a clinician respond when a patient's fear returns after successful treatment?

Research assignments

- Design a conditioning study that tests generalisation in PTSD, specifying comparison groups and measures.
- Critically appraise one trial of exposure therapy, noting the mechanism it claims to test.
- Write a session-by-session exposure plan for a specific fear, including expectancy violation and context variation.

Chapter summary

Threat learning is the process by which neutral cues acquire the capacity to elicit defensive responses. It depends on amygdala plasticity, is modulated by stress hormones and context, and is maintained by avoidance. Extinction is new inhibitory learning, not erasure, and is subject to relapse. Prediction error determines how much is learned. These principles inform exposure therapy and explain both its effectiveness and its limitations.

References

Craske, M. G., Treanor, M., Conway, C. C., Zbozinek, T., & Vervliet, B. (2014). Maximizing exposure therapy: An inhibitory learning approach. *Behaviour Research and Therapy, 58*, 10–23. https://doi.org/10.1016/j.brat.2014.04.006

Ehlers, A., & Clark, D. M. (2000). A cognitive model of posttraumatic stress disorder. *Behaviour Research and Therapy, 38*(4), 319–345. https://doi.org/10.1016/S0005-7967(99)00123-0

Fenster, R. J., Lebois, L. A. M., Ressler, K. J., & Suh, J. (2018). Brain circuit dysfunction in post-traumatic stress disorder: From mouse to man. *Nature Reviews Neuroscience, 19*(9), 535–551. https://doi.org/10.1038/s41583-018-0039-7

LeDoux, J. E. (2000). Emotion circuits in the brain. *Annual Review of Neuroscience, 23*, 155–184. https://doi.org/10.1146/annurev.neuro.23.1.155

Litz, B. T., Stein, N., Delaney, E., Lebowitz, L., Nash, W. P., Silva, C., & Maguen, S. (2009). Moral injury and moral repair in war veterans: A preliminary model and intervention strategy. *Clinical Psychology Review, 29*(8), 695–706. https://doi.org/10.1016/j.cpr.2009.07.003

Maren, S. (2001). Neurobiology of Pavlovian fear conditioning. *Annual Review of Neuroscience, 24*, 897–931. https://doi.org/10.1146/annurev.neuro.24.1.897

Milad, M. R., & Quirk, G. J. (2012). Fear extinction as a model for translational neuroscience: Ten years of progress. *Annual Review of Psychology, 63*, 129–151. https://doi.org/10.1146/annurev.psych.121208.131631